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Experiment: Vacuum Cleaner Problem

Aim

To implement the Vacuum Cleaner Problem using Python and demonstrate the working of a simple reflex agent in Artificial Intelligence.

Objectives

- To understand the concept of a Vacuum Cleaner Agent.
- To implement a simple reflex agent using Python.
- To clean all dirty rooms based on the current environment.
- To understand the working of intelligent agents in AI.
- To analyze the performance of the vacuum cleaner agent.

Algorithm

1. Initialize the environment with two rooms (A and B).
2. Check the current room's status (Dirty or Clean).
3. If the room is dirty, clean it.
4. Move to the next room.
5. Repeat the process until all rooms are clean.
6. Display the status before and after cleaning.

Python Program

Vacuum Cleaner Problem

```
def vacuum_cleaner(room):  
    print("Initial Room Status:")  
    print(room)
```

```
for location in room:
    if room[location] == "Dirty":
        print("Cleaning Room", location)
        room[location] = "Clean"
    else:
        print("Room", location, "is already Clean")
```

```
print("\nFinal Room Status:")
print(room)
```

Driver Code

```
rooms = {
    "A": "Dirty",
    "B": "Dirty"
}
```

```
vacuum_cleaner(rooms)
```

Sample Output

Initial Room Status:

```
{'A': 'Dirty', 'B': 'Dirty'}
```

Cleaning Room A

Cleaning Room B

Final Room Status:

```
{'A': 'Clean', 'B': 'Clean'}
```

Out put

<pre>1 # Vacuum Cleaner Problem 2 3 def vacuum_cleaner(room): 4 print("Initial Room Status:") 5 print(room) 6 7 for location in room: 8 if room[location] == "Dirty": 9 print("Cleaning Room", location) 10 room[location] = "Clean" 11 else: 12 print("Room", location, "is already Clean") 13 14 print("\nFinal Room Status:") 15 print(room) 16 17 # Driver Code 18 rooms = { 19 "A": "Dirty", 20 "B": "Dirty" 21 } 22 23 vacuum_cleaner(rooms)</pre>	<pre>Initial Room Status: {'A': 'Dirty', 'B': 'Dirty'} Cleaning Room A Cleaning Room B Final Room Status: {'A': 'Clean', 'B': 'Clean'} === Code Execution Successful ===</pre>
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Result

The Vacuum Cleaner Problem was successfully implemented using Python. The vacuum cleaner agent cleaned all dirty rooms and updated the environment to the clean state.

Conclusion

The experiment demonstrated the working of a **Simple Reflex Agent** in Artificial Intelligence. The vacuum cleaner agent sensed the state of each room and performed the appropriate action (cleaning if dirty). This experiment helps in understanding the fundamentals of intelligent agents and their decision-making process in AI.